Övning 4. Termoelasticitet o plasticitet

2.1:36 2.2:20, 30, 31, 33

Linjärt termoelastiskt material

$$\varepsilon = \frac{\sigma}{E} + \alpha \Delta T \qquad [FS\ 3.1] \quad (\delta = \varepsilon L)$$

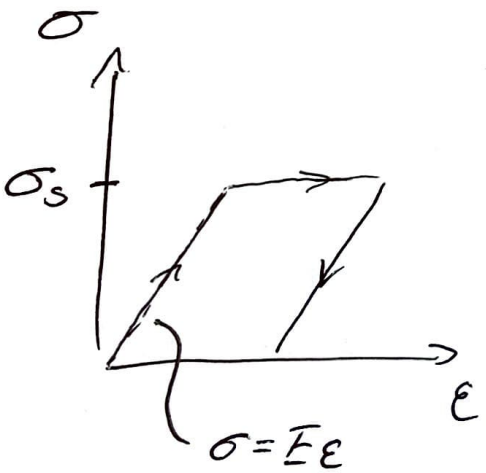
$\frac{\sigma}{E}$ ↑ Mekanisk töjning

$\alpha \Delta T$ ↑ Termisk töjning

α - längdutvidgnings koefficient

ΔT - Temperatur förändring

Elastiskt - idealplastiskt material

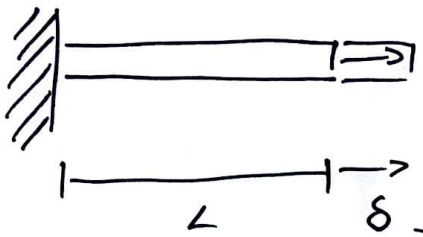


σ_s - Sträckgräns (Material egenskap)

P_s - Plasticeringslast, Kraft för initiell plasticering

P_f - Kollapslast, Kraft då alla delar uppnår sträckgränsen.

Ex:



ΔT förändring

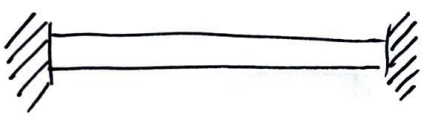
$$\varepsilon = \frac{\sigma}{E} + \alpha \Delta T$$

$\sigma = 0$

$$\varepsilon = \alpha \Delta T$$

$$\delta = \varepsilon L = \alpha \Delta T L$$

Fast inspänd



ΔT förändring

$$\varepsilon = \frac{\sigma}{E} + \alpha \Delta T$$

$$\delta = 0 \Rightarrow \varepsilon = 0$$

$$\frac{\sigma}{E} = -\alpha \Delta T \Rightarrow \sigma = -\alpha \Delta T E$$

Positiv temperatur förändring $\sigma < 0$ komp

Negativ temperatur förändring $\sigma > 0$ Drag

Ex: Plasticitet





$$\sigma = \frac{P}{A}$$

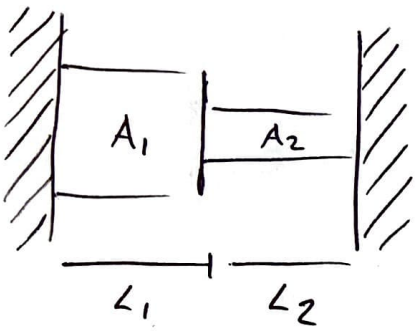
$$\sigma = \frac{P}{2A}$$

Stång 1 kommer plasticera först

2.1:36

Givet: $A_1, A_2, L_1, L_2, E, \alpha, \Delta T$

Sökt: Spänningarna i båda stängerna σ_1, σ_2



① Frilägging

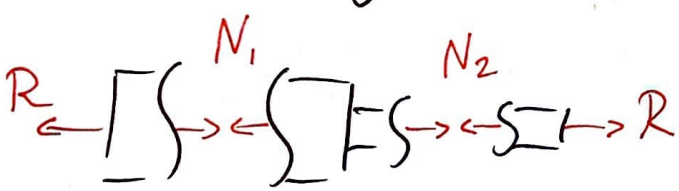


② Jämvikt

$\rightarrow: \; R_2 - R_1 = 0$

$R_1 = R_2 = R$

③ Snittning



④ Jämvikt

$\text{II} \rightarrow: \; N_2 - N_1 = 0$

$$\boxed{N_1 = N_2 = R}$$

$\sigma_1 A_1 = \sigma_2 A_2 \Rightarrow \sigma_1 = \sigma_2 \frac{A_2}{A_1}$

⑤ Deformationssamband

$\delta_{tot} = 0 \qquad \delta_1 + \delta_2 = 0 \qquad \delta_i = \varepsilon_i L$

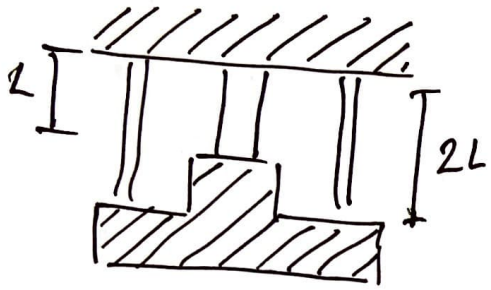
$$\Rightarrow \left(\frac{\sigma_1}{E} + \alpha\Delta T\right) L_1 + \left(\frac{\sigma_2}{E} + \alpha\Delta T\right) L_2 = 0$$

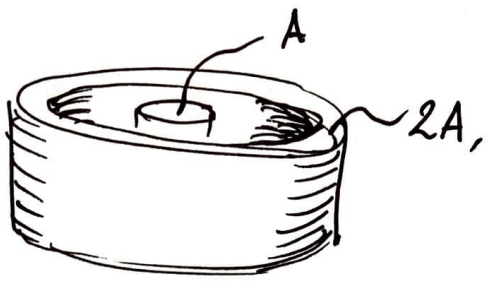
$$\sigma_1 \left(\frac{L_1}{E} + \frac{A_1}{A_2}\frac{L_2}{E}\right) + \alpha\Delta T (L_1 + L_2) = 0$$

$$\Rightarrow \boxed{\sigma_1 = -\frac{\alpha\Delta T E (L_1 + L_2)}{L_1 + L_2 \frac{A_1}{A_2}}} \qquad \boxed{\sigma_2 = -\frac{\alpha\Delta T E (L_1 + L_2)}{\frac{A_2}{A_1} L_1 + L_2}}$$

2.2.20

$A, E, L, \alpha, \Delta T$





Sökt: Temperaturökningen orsakar spänning i stav och rör. Vilken?

σ_{stav} , $\sigma_{rör}$

① 

$N_S + N_R = 0$

Snittning

② Deformationssamband

$\delta_R = \delta_S$

$\delta_i = \left(\frac{N_i}{EA_i} + \alpha\Delta T\right) L_i$

$$\left(\frac{N_R}{2EA} + \alpha\Delta T\right) 2L = \left(\frac{N_S}{EA} + \alpha\Delta T\right) L$$

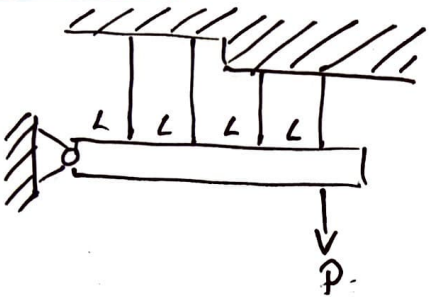
$$\Rightarrow N_R + EA\alpha\Delta T = N_S$$

$$N_S = \frac{1}{2} EA\alpha\Delta T \qquad N_R = -\frac{1}{2} EA\alpha\Delta T$$

Röret i tryck Stången i drag

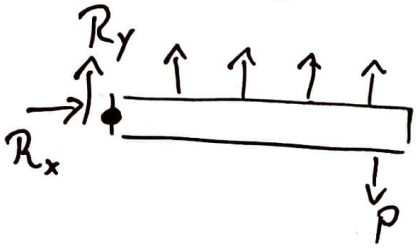
$$\sigma_S = \frac{1}{2} E\alpha\Delta T \qquad \sigma_R = -\frac{1}{4} E\alpha\Delta T$$

2.2.30



Givet: L, A,
σ_s - Sträckgränsen
Material egenskap
Sökt: Kollapslast P_f

① Frilägging



Jämvikt

$\uparrow$: $R_y + N_1 + N_2 + N_3 + N_4 - P = 0$ (1)

$\rightarrow$: $R_x = 0$

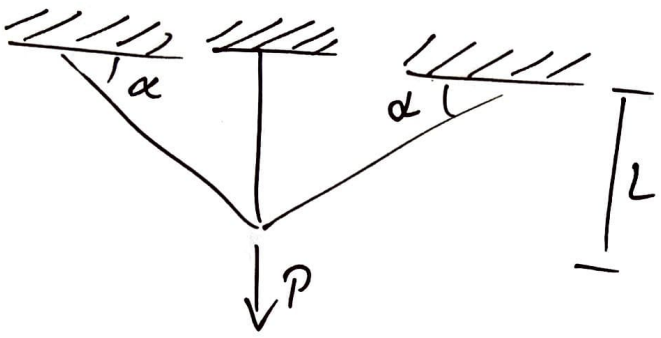
$\circlearrowleft$: $N_1 L + N_2 2L + N_3 3L + N_4 4L - P 4L = 0$ (2)

Vid kollapslast uppnår alla stänger plasticering

$N_1 = N_2 = N_3 = N_4 = \sigma_s A$ $\left(\sigma_s = \frac{N}{A}\right)$

$$\begin{cases} (1): & 4\sigma_s A + R_y = P_f \\ (2): & 10\sigma_s A = 4P_f \end{cases} \Rightarrow \boxed{P_f = \frac{5}{2}\sigma_s A}$$

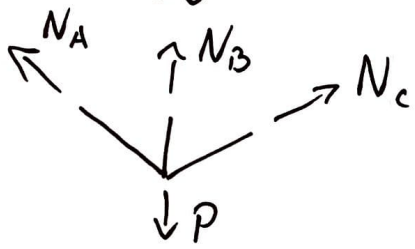
2.2.31



Givet: E, A, L, α

Sökt: Kollapslast P_f

① Frilägg



Jämvikt

$\rightarrow$: $N_A \cos\alpha = N_C \cos\alpha$

$N_A = N_C$

$\uparrow$: $N_B + 2N_A \sin\alpha - P = 0$ (1)

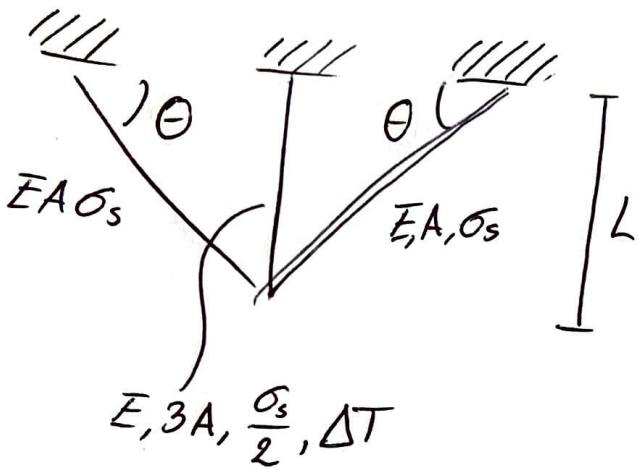
Kollaps fås då alla stänger plasticerat

$N_A = N_B = \sigma_s A$

(1) $\Rightarrow \sigma_s A + 2\sigma_s A \sin\alpha - P_f = 0$

$$\boxed{P_f = \sigma_s A (1 + 2\sin\alpha)}$$

2.2:33

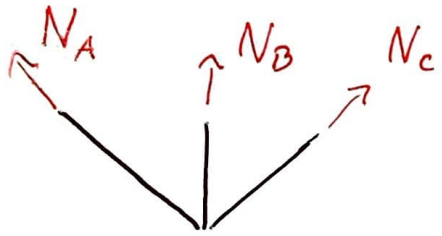


Givet: E, A, σ_s $\quad \theta = 45°$

Mittstången värms upp ΔT

Sökt: ΔT för inledande plasticering

① Frilägging



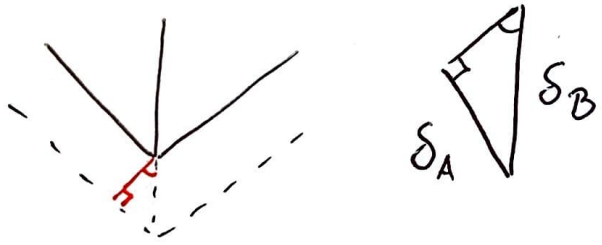
② Jämvikt

$\rightarrow: \; N_A = N_C$

$\uparrow: \; N_B + N_A \sin\theta + N_C \sin\theta = 0$

$\Rightarrow N_B = -\sqrt{2}\, N_A$

③ Deformationssamband



$$\sin\theta = \frac{\delta_A}{\delta_B} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow \delta_A = \frac{1}{\sqrt{2}}\delta_B$$

$$\delta_A = \frac{N_A \sqrt{2} L}{EA} \qquad \delta_B = \frac{N_B L}{E 3A} + \alpha \Delta T L$$

$$\Rightarrow \frac{N_A \sqrt{2} L}{EA} = \frac{1}{\sqrt{2}}\left(\frac{N_B L}{3EA} + \alpha \Delta T L\right)$$

$$\Rightarrow N_A = \frac{3}{6+\sqrt{2}} EA\alpha\Delta T \simeq 0.4\, EA\alpha\Delta T$$

$$N_B = -\frac{3\sqrt{2}}{6+\sqrt{2}} EA\alpha\Delta T \simeq -0.57\, EA\alpha\Delta T$$

2.2:33 del 2

④ Plasticering i de olika stängerna

$$\sigma_A = \sigma_S = \frac{N_A}{A} = \frac{3}{6+\sqrt{2}} \frac{EA}{A} \alpha \Delta T_{SA}$$

$$\Rightarrow \Delta T_{SA} = \frac{6+\sqrt{2}}{3} \frac{\sigma_S}{E\alpha} \approx 2.47 \frac{\sigma_S}{E\alpha}$$

$$\sigma_B = \frac{\sigma_S}{2} = \frac{N_B}{3A} = -\frac{\sqrt{2}}{6+\sqrt{2}} E\alpha \Delta T_{SB}$$

$$\Rightarrow \Delta T_{SB} = -\frac{6+\sqrt{2}}{2\sqrt{2}} \frac{\sigma_S}{E} \approx -2.62 \frac{\sigma_S}{E\alpha}$$

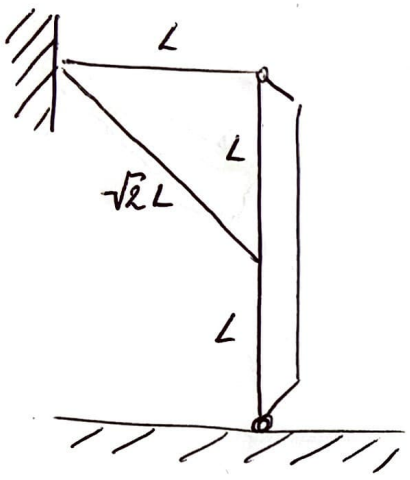
Stång A/C kräver lägre temp förändring för att plasticering skall ske.

$$\Delta T_S = \max\left(|\Delta T_{SA}|, |\Delta T_{SB}|\right) = 2.47 \frac{\sigma_S}{E\alpha}$$

Enhetskoll

$$\left[\frac{N}{m^2} \Big/ \left(\frac{N}{m^2} \cdot \frac{1}{^\circ K}\right)\right] = [^\circ K]$$

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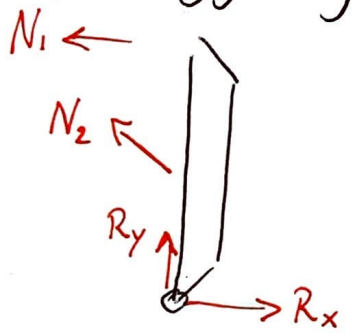


Givet: $d, E, L, \alpha, \Delta T$

Beräkna inom vilket intervall ΔT måste vara så ingen av stängerna får en kompressiv belastning som överstiger euler 2.

$$N_{1,min} = -\frac{\pi E A^2}{L^2} \qquad N_{2,min} = -\frac{\pi E A^2}{L^2}$$

① Frilägging



② Jämvikt

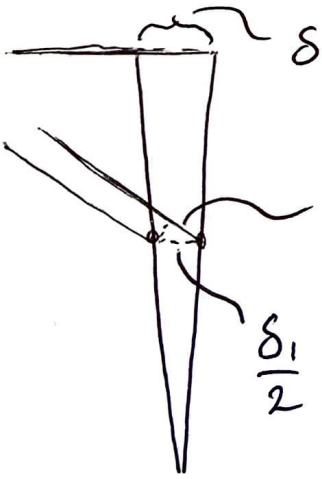
$$\rightarrow: -N_1 - N_2 \frac{1}{\sqrt{2}} + R_x = 0$$

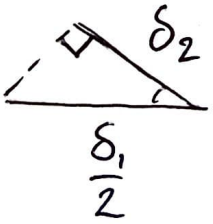
$$\uparrow: N_2 \frac{1}{\sqrt{2}} + R_y = 0$$

$$\circlearrowright: -N_1 2L - N_2 \frac{1}{\sqrt{2}} L = 0$$

4 obekanta
3 ekvationer

③ Deformation





$$\frac{1}{\sqrt{2}} = \frac{\delta_2}{\frac{\delta_1}{2}}$$

$$\Rightarrow \delta_1 = 2\sqrt{2}\,\delta_2$$

$$\frac{N_1 L}{EA} + \alpha \Delta T L = 2\sqrt{2}\left(\frac{N_2 \sqrt{2} L}{EA} + \alpha \Delta T \sqrt{2} L\right)$$

$$N_1\left(\frac{L}{EA} + 8\sqrt{2}\frac{L}{EA}\right) = \alpha \Delta T L (4-1)$$

$$\Rightarrow \quad N_1 = \frac{3\alpha \Delta T}{1+8\sqrt{2}} EA$$

$$N_2 = -\frac{6\sqrt{2}\alpha \Delta T}{1+8\sqrt{2}} EA$$

Positiva ΔT N_2 i kompression

Negativa ΔT N_1 i kompression

$$N_1 = -\frac{\pi E A^2}{2L^2} = \frac{3\alpha \Delta T_I}{1+8\sqrt{2}} EA$$

$$\Rightarrow \boxed{\Delta T_I = -\frac{(1+8\sqrt{2})\pi A}{6\alpha L^2}}$$

$$N_2 = -\frac{\pi E A^2}{4L^2} = -\frac{6\sqrt{2}\alpha \Delta T}{1+8\sqrt{2}} EA$$

$$\Rightarrow \boxed{\Delta T_{II} = \frac{(1+8\sqrt{2})\pi A}{24\sqrt{2}\,\alpha L^2}}$$